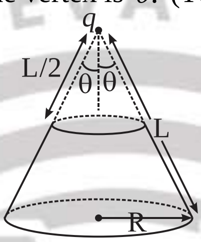
Time Allowed: 1 : 30 Hrs

PHYSICS

Instructions (Q 1 to Q 5): Multiple Option Correct type MCQs. (One or More than One correct)

1. A charge q is located at the tip of a hollow cone with surface charge density σ. The slant height of the cone is L, and the half angle at the vertex is θ. (Top half of the cone is removed as shown.)



Choose the correct option(s)

(A) The magnetic of force on charge 'q' due to the cone is $\dfrac{q\sigma \sin 2\theta}{4\varepsilon_0}\ell n(2)$

(B) The magnetic of force on charge 'q' due to the cone is $\dfrac{q\sigma \cos 2\theta}{2\varepsilon_0}\ell n(2)$

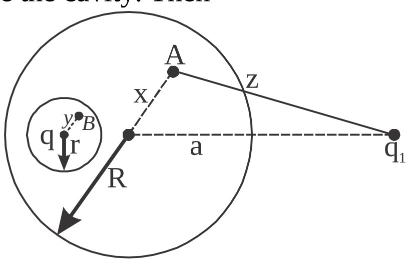
(C) Force on charge 'q' will be maximum for, $\theta = 90^o$

(D) Force on charge 'q' will be maximum for, $\theta = 45^o$

2. A vessel is partly filled with liquid. When the vessel is cooled to a lower temperature, the space in the vessel unoccupied by the liquid remains constant. Then the volume of the liquid (V_L), volume of the vessel (V_v), the coefficients of cubical expansion of the material of the vessel (γ_v) and of the liquid (γ_L) are related as

(A) $\gamma_L > \gamma_v$ (B) $\gamma_L < \gamma_v$ (C) $\gamma_v / \gamma_L = V_v / V_L$ (D) $\gamma_v / \gamma_L = V_L / V_v$

3. A cavity is made inside a solid conducting sphere and a charge q is placed inside the cavity at the centre. A charge q_1 is placed outside the sphere as shown in the figure. Point A is inside the sphere and point B is inside the cavity. Then



(A) Electric field at point A is zero.

(B) Electric field at point B is $\dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{y^2}$

(C) Potential at point A is $\dfrac{1}{4\pi\varepsilon_0}\left[\dfrac{q_1}{a}+\dfrac{q}{R}\right]$

(D) Potential at point B is $\dfrac{1}{4\pi\varepsilon_0}\left[\dfrac{q_1}{a}+\dfrac{q}{R}+\dfrac{q}{y}-\dfrac{q}{r}\right]$

4. A soap bubble of radius 'r' and wall thickness 'h' is made in vacuum by blowing an ideal diatomic gas in it. The surface tension of the soap solution is 'σ' and density of the soap solution is 'ρ'. Assume that the heat capacity of the soap film is much greater than that of the gas in the bubble. Then choose the correct option(s). ($R =$ ideal gas constant)

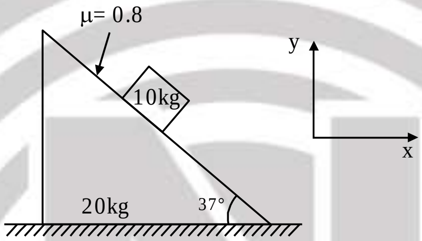
(A) The molar heat capacity of the gas in the bubble is 4R.

(B) The molar heat capacity of the gas in the bubble is 3R.

(C) The time period of small radial oscillations of the soap film is $2\pi\sqrt{\dfrac{\rho h r^2}{8\sigma}}$

(D) The time period of small radial oscillations of the soap film is $2\pi\sqrt{\dfrac{\rho h r^2}{4\sigma}}$

5. As shown in the figure a wedge of mass 20 kg is moving rightwards on which a block of mass 10 kg is placed on it. Friction coefficient between the wedge and the block is 0.8. [take g = 10 m/s^2]. Select correct alternative(s) among the following options.



(A) If wedge is moving with constant velocity then normal force on wedge from the ground is 300N

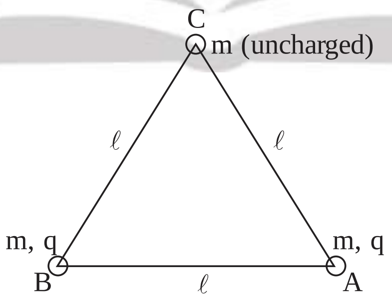
(B) If wedge is moving with constant velocity then acceleration of block is zero

(C) If wedge is moving with $\vec{a} = 2(\hat{i})\, m/s^2$ then friction acting on block is 44N

(D) If wedge is moving with $\vec{a} = 10(\hat{i})\, m/s^2$ then friction is 20 N down the inclined plane on the wedge

Instructions (Q6 to Q10): Numerical Value Answer type Questions.

6. System is placed in Horizontal plane.



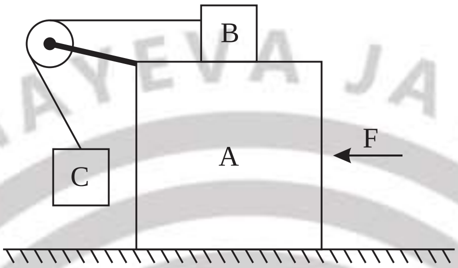
If string between A and B is cut, maximum speed of c during the motion is $\sqrt{\dfrac{kq^2}{xml}}$. find value of x.

7. On a frosty day, Mr. Thomas grab some hot chocolate from roadside shop. After paying, Mr. Thomas drop 9 gram of copper coin (initially at $-12.0^\circ C$) into his coat pocket. Inside that Pocket are already 14 gram of silver coin at a much warmer $30^\circ C$. After some time, the copper coin reach $4^\circ C$. Find out specific heat of silver in nearest integer (in $J/kg-^\circ C$)

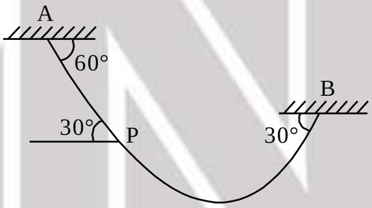
(specific heat of copper $= 387\, J/kg^\circ C$)

8. A thin uniform rod of length $l = 1\text{m}$ and mass $m = 8\text{gm}$ is free to rotate about a horizontal axis passing through one of its ends. The rod is released from horizontal position in vertical plane. Find the shear stress (in N/m^2) developed at the centre of rod immediately after it is released. Area of cross-section of rod $A = 1.0\,\text{cm}^2$ and for calculation of Moment of inertia you can treat it to be very thin. $\left(g = 10\,\text{m}/\text{s}^2\right)$

9. In the shown arrangement, a force *F* moves the blocks on smooth horizontal surface. Blocks *B* and *C* are at rest relative to block *A* without contact between *C* & *A*. All surfaces are smooth.
Find the value of '*F*' (in *N*), if $m_A = 12\,\text{kg}$, $m_B = 5\text{kg}$ and $m_c = 3kg \cdot \left(g = 10\,m/s^2\right)$



10. A uniform chain of mass M = 10 kg hangs in vertical plane as shown is figure. Find mass of segment AP of the chain. [Take g = 10 m/s^2]



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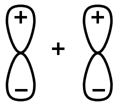
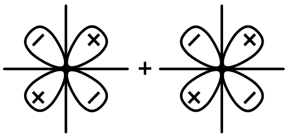
CHEMISTRY

Instructions (Q 1 to Q4): Multiple Option Correct type MCQs. (One or More than One correct)

1. Which of the following overlapping result ungerade molecular orbital?

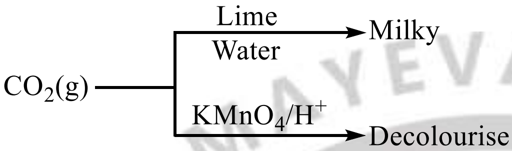
(A) 

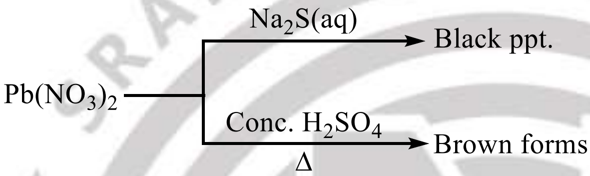
(B) 

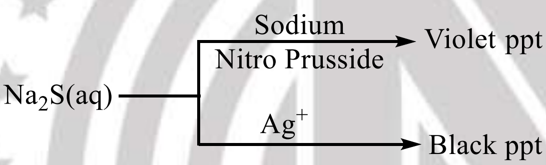
(C) 

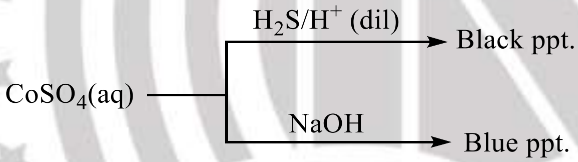
(D) 

2. Choose the correct test of species among given

(A) 

(B) 

(C) 

(D) 

3. Which of the following comparison(s) represent the correct order of properties indicated against it?

(A) P > F > Si (Covalent radii)

(B) Ne > Ar > He (Electron Gain Enthalpy with sign)

(C) H > Li > Na (Ionization energy)

(D) Cl > O > Ne (Electron affinity)

4. How many of the following reactions produce blue colouration or blue ppt.

(A) $CuSO_4$ solution + *NH_3 in excess* $\longrightarrow$

(B) $CuSO_4$ solution + $K_4[Fe(CN)_6]$ $\longrightarrow$

(C) $NiCl_2$ solution + *Excess NH_3* $\longrightarrow$

(D) $FeSO_4$ solution + $K_3[Fe(CN)_6]$ $\longrightarrow$

Instructions (Q5 to Q10): Numerical Value Answer type Questions.

5. $O + e^- \rightarrow O^-$

$Be^+ + e^- \rightarrow Be$

$N + e^- \rightarrow N^-$

$S^- + e^- \rightarrow S^{-2}$

$P^- \rightarrow P + e^-$

$Ne^+ + e^- \rightarrow Ne$

$O^- + e^- \rightarrow O^{-2}$

$Mg + e^- \rightarrow Mg^-$

$Cl^- \rightarrow Cl + e^-$

$S^- \rightarrow S + e^-$

$O + 2e^- \rightarrow O^{-2}$

$Ne^- \rightarrow Ne + e^-$

$Na^+ + e^- \rightarrow Na$

If x = number of endothermic reactions and

y = number of reactions having a product with noble gas configuration

Then calculate (x + y)

6. Calculate the maximum number of atoms lying in one plane of CH_2SF_4

7. Consider following reactions:

(i). $CuSO_4 + KCN \rightarrow$ products

(ii). NO_3^{-1} + cons. $H_2SO_4 + FeSO_4 \rightarrow$ products

(iii). $Fe^{2+} + [Fe(CN)_6]^{4-} \xrightarrow[\text{of air}]{\text{absence}}$ products

(iv). $Fe(SCN)_3 + C_2O_4^{2-} \rightarrow$ products

(v) $Co^{3+} + aq.NH_3 \longrightarrow$ products

In how many of the reactions, the product contains paramagnetic central metal ion ?

8. Number of dibasic OXO acids that contain bond between two identical atoms (X – X)

$H_2S_2O_8,\ H_4P_2O_5,\ H_5P_3O_{10},\ H_2S_4O_6,\ H_4P_2O_8,\ H_2N_2O_2$

9. How many of the following are square planar ?

(A) $[Ni(dmg)_2]$ (B) $[Ni(CN)_4]^{2-}$ (C) $[Ni(CN)_4]^{4-}$

(D) $[Ni(CO)_4]$ (E) $[Ni(Cl)_4]^{2-}$ (F) NF_4^+

(G) K_2CrO_4 (H) $KMnO_4$ (I) K_2MnO_4

(J) CrO_2Cl_2

10. Consider the complex $[Pt(S_5)_3]^{2-}$. If x, y and z are the number of chelates, number of optically active isomers and number of geometrical isomers respectively, find the value of $(x+y+z)$:

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MATHS

Instructions (Q 1 to Q3): Multiple Option Correct type MCQs. (One or More than One correct)

1. Let $N = 10^{3\log 2 - 2\log\left(\log 10^3\right) + \log\left(\left(\log 10^6\right)^2\right)}$ where base of logarithm is 10.

If N^{N^N} is divided by 7, we get the remainder R. Which of the following options are correct?

(A) R is perfect square of natural number.

(B) R is prime number.

(C) R is even number

(D) R is factor of 20.

2. The equation of straight line with gradient 1, passing through $\left(\frac{m}{2}, \frac{n}{2}\right)$ where $m, n \in R$ satisfies the equation $\sec^2(n(m+2)) + m^2 = 1$ $\left(\text{where } n \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]\right)$ can be

(A) $x + y = 0$

(B) $x - y = 0$

(C) $x - y + \frac{\pi}{4} = 0$

(D) $x - y - \frac{\pi}{4} = 0$

3. If S denotes the sum of reciprocal of all those natural numbers whose prime factors can be only 2, 3 or 5, then the value of 4S is less than

(A) 15

(B) 17

(C) 19

(D) 21

Instructions (Q4 to Q10): Numerical Value Answer type Questions.

4. If the minimum value of $|x| - |y|$ when

$\log_4(x-2y) + \log_4(x+2y) - 1 = 0$

is $\sqrt{k}$, where k is a prime, then the value of k^2 is:

5. If x, y, z are positive reals such that $x^2 + y^2 + z^2 = 7; xy + yz + zx = 4$ then the minimum value of xy is $\frac{p}{q}$ (p, q are relatively prime). Find the value of ($p + q$).

6. The sequence a_n and b_n for $n \in N$ are defined as $a_n = 3n + \sqrt{n^2 - 1}$ and $b_n = 2\left(\sqrt{n^2 + n} + \sqrt{n^2 - n}\right)$.

If $\sum_{n=1}^{49} \sqrt{a_n - b_n} = -a + b\sqrt{c}$ where $a, b, c \in N$ and $b + c$ is minimum, then find the number of divisors of $N = a^{29} \cdot b^{23} \cdot c^{20}$

7. Let T_n be the n^{th} term of Series for $n \geq 1$ and $T_1 = 1, T_{n+1} = 2T_n + 4^n, n \geq 1$. If $S_n =$ Sum of first n terms of the series and $S_{2025} = \frac{a \cdot 2^{4050} + b \cdot 2^{2025} + 1}{3}$ where $a, b \in I$, then absolute value of $\frac{2b}{a}$ is

8. Let A and B be two sets containing 3 elements and 3 elements respectively. The number of subsets of (A × B) having 3 or more elements is

9. Let $X = \{(a,b) : a,b \in I, a \geq 0, b \geq 0 \text{ and } 4a + 5b \leq 40\}$

$Y = \{(a,b) : a,b \in I, a \geq 0, b \geq 0 \text{ and } 5a + 4b \leq 40\}$

Where I denotes set of integers, then number of elements in $X \cap Y$ is ____

10. If $3\alpha_1 + 4\beta_1 = 10$ and $4\alpha_2 - 3\beta_2 + 20 = 0$ and let (p, q) is the orthocenter and Δ is the area of triangle formed by $(-2, 4)$, (α_1, β_1) and (α_2, β_2) where $\alpha_i^2 + \beta_i^2$ is minimum $(\forall i = 1, 2)$ then find the value of $p^2 + q^2 + \Delta^2$.

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ANSWER KEY

PHYSICS									
1.	**2.**	**3.**	**4.**	**5.**	**6.**	**7.**	**8.**	**9.**	**10.**
AD	**AD**	**ABCD**	**AC**	**ABCD**	**6**	**153**	**50**	**150**	**5**
CHEMISTRY									
1.	**2.**	**3.**	**4.**	**5.**	**6.**	**7.**	**8.**	**9.**	**10.**
AD	**BD**	**BCD**	**ACD**	**13**	**6**	**2**	**4**	**2**	**5**
MATHS									
1.	**2.**	**3.**	**4.**	**5.**	**6.**	**7.**	**8.**	**9.**	**10.**
ACD	**BCD**	**BCD**	**9**	**5**	**2010**	**3**	**466**	**45**	**36**